Software Engineering

Chapter 6: User Experience (UX) Engineering & Interaction Design

From Cognitive Foundations & Quantitative Laws to Empirical Usability Engineering

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Chapter 6: Roadmap & Core Curriculum

Part 1: Foundations, Cognition, Laws & Research

- **6.1 Foundations of HCI, UX & Usability:** Engelbart's GUI revolution, ISO 9241, Garrett's 5 Planes, and Nielsen's 5 usability dimensions.
- **6.2 Cognitive Psychology & Mental Models:** Don Norman's 7 Stages of Action, the Two Gulfs, Signifiers, Mental Models, and Cognitive Load.
- **6.3 Quantitative Laws & Visual Perception:** Ben Shneiderman's Direct Manipulation, Fitts's Law, Hick's Law, Gestalt grouping, and WCAG a11y.
- **6.4 Human-Centered Design & User Research:** Double Diamond lifecycle, 2×2 Research Matrix, Personas, JTBD, and Customer Journey Maps.

Part 2: Architecture, Heuristics, Testing & AI UX

- **6.5 Information Architecture & Prototyping:** LATCH taxonomy, Card Sorting, Tree Testing, Wireframing, and Brad Frost's Atomic Design.
- **6.6 Analytical Evaluation (Heuristics):** Jakob Nielsen's 10 Usability Heuristics, 0–4 Severity Scale, and Cognitive Walkthroughs.
- **6.7 Empirical Testing, Metrics & Ethics:** Herbert Simon's Think-Aloud protocol, N=5 Poisson curve, SUS 0–100, HEART, and Dark Patterns.
- **6.8 AI-Augmented UX & Synthesis:** AI for UX vs. UX for AI (Microsoft 18 HAI Guidelines), interactive recap quiz, and reference solutions.

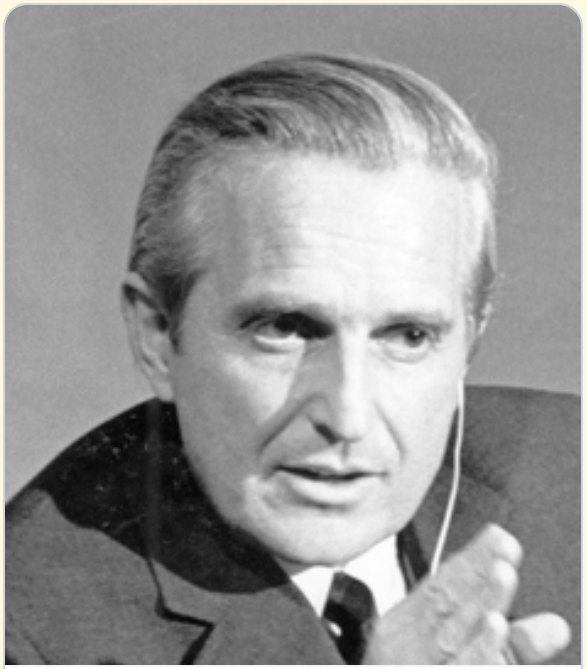
6.1 Foundations of HCI, UX & Usability Engineering

> *"Design is not just what it looks like and feels like. Design is how it works."*
> *— Steve Jobs (Co-founder of Apple)*

Pioneers of HCI: Douglas Engelbart & Interactive Computing

- **Role & Distinctions:**
  - Director, Augmentation Research Center (SRI International); **1997 ACM Turing Award Laureate**; National Medal of Technology.
- **The 1968 "*Mother of All Demos*" (San Francisco):**
  - When computing meant batch punch cards, Engelbart demonstrated live **bitmapped windows, hypertext links, real-time collaborative editing, video conferencing**, and his invention: the **Computer Mouse**.
- **Core Philosophy — Augmenting Human Intellect:**
  - Viewed software not as an isolated number-cruncher, but as a **co-evolving human-machine system** designed to amplify human cognitive capability.
- **Foundational HCI Maxim:**

> *"The digital revolution is far more significant than the invention of writing or even of printing—because it augments collective human intellect."*



Douglas Engelbart (1925–2013)
SRI International / ACM Turing Award

What is User Experience (UX)? Formal Definitions

> ***ISO 9241-210 Definition:*** *User Experience is "a person's perceptions and responses resulting from the use and/or anticipated use of a system, product or service."*

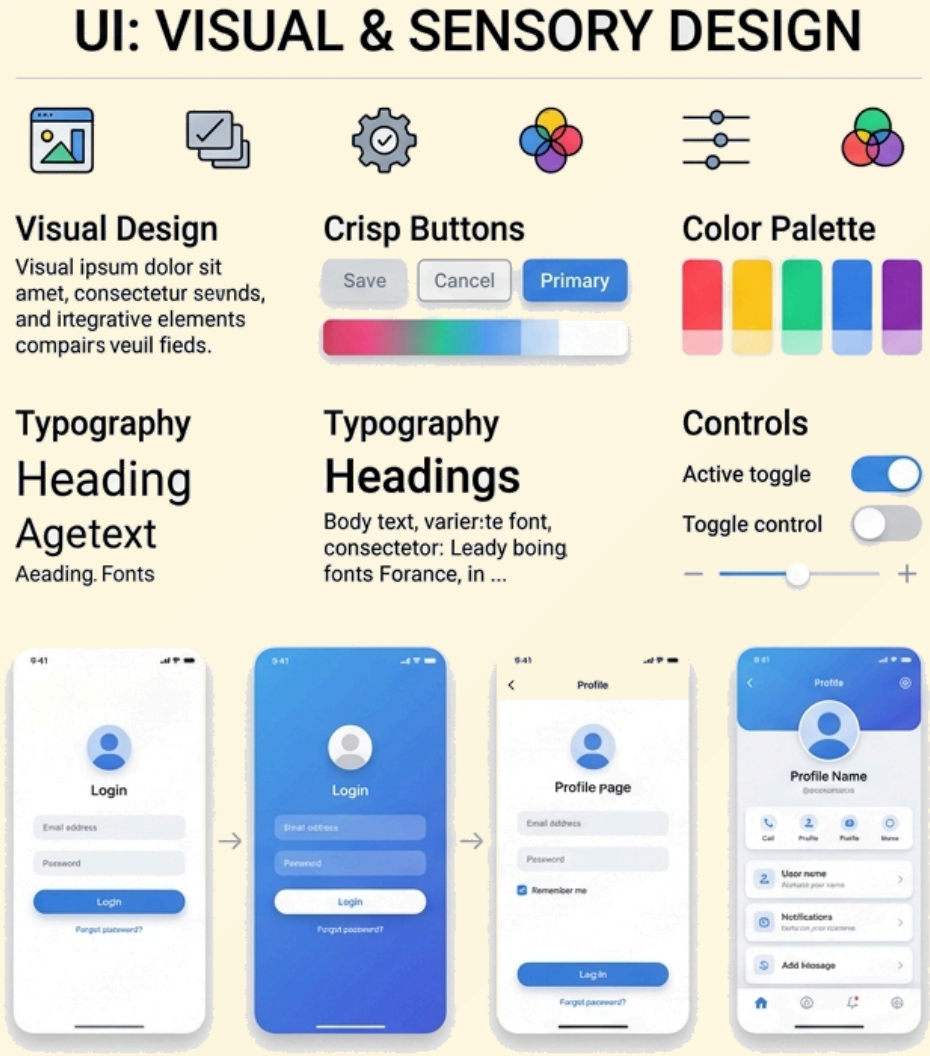
1. HCI vs. UX vs. UI vs. CX

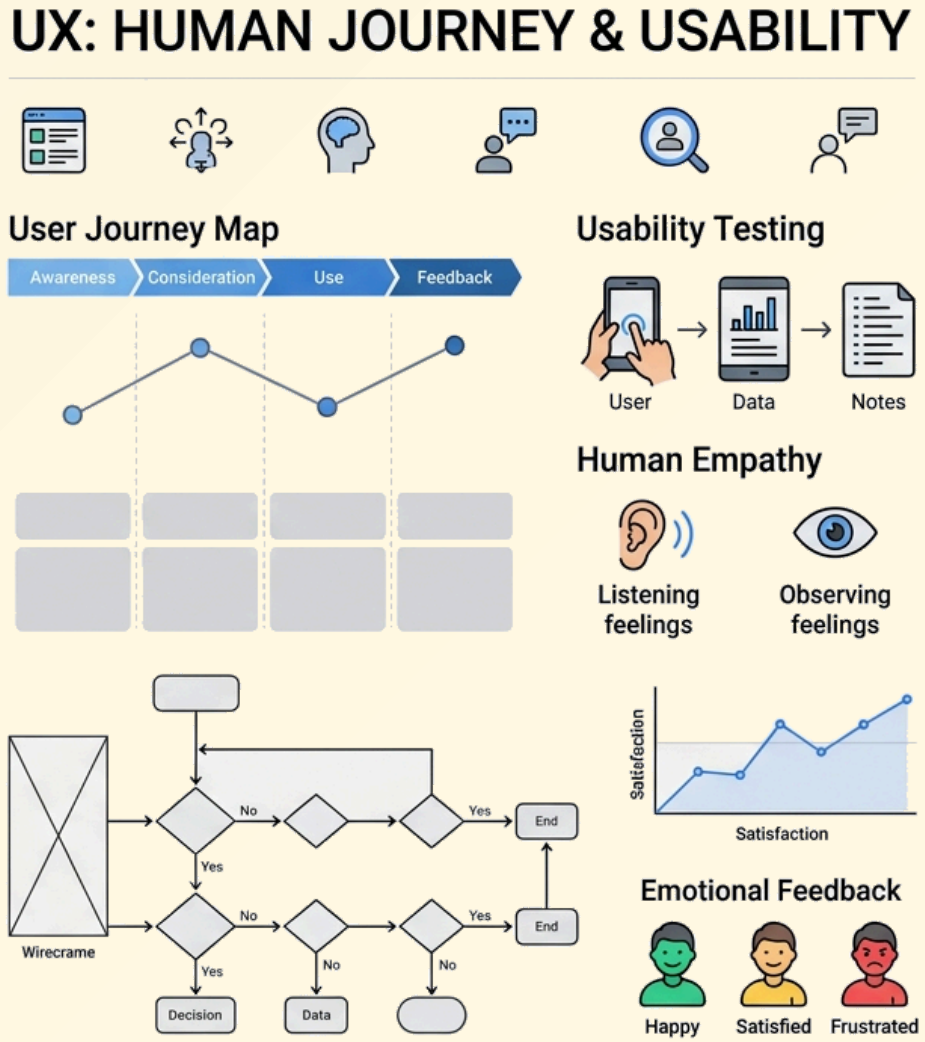
- **Human-Computer Interaction (HCI)**:
  - The academic discipline uniting computer science, cognitive psychology, and ergonomics to study interactive computing systems.
- **User Interface (UI — The Conduit)**:
  - The concrete visual, auditory, or tactile presentation layer (buttons, typography, layouts, voice prompts) through which users interact.
- **User Experience (UX — The Holistic Outcome)**:
  - The end-to-end cognitive, behavioral, and emotional journey before, during, and after interacting with the system.
- **Customer Experience (CX)**:
  - The broader organizational umbrella encompassing marketing, pricing, retail support, and multi-channel brand interactions.

2. Why UI Alone Cannot Save a Broken UX

- **[Case Study] The Food Delivery Checkout Paradox**:
  - Suppose a Food Delivery app features award-winning gradients and smooth 120Hz animations (**Great UI**).
  - However, it hides delivery surge fees until the final step, loses the cart when switching payment methods, and offers no cancel button (**Broken UX**).
- **Engineering Insight**:
  - **UI** is the saddle, the stirrups, and the reins; **UX** is whether the rider actually reaches their destination safely and comfortably.
  - A beautiful UI on top of a flawed task workflow only accelerates user frustration.

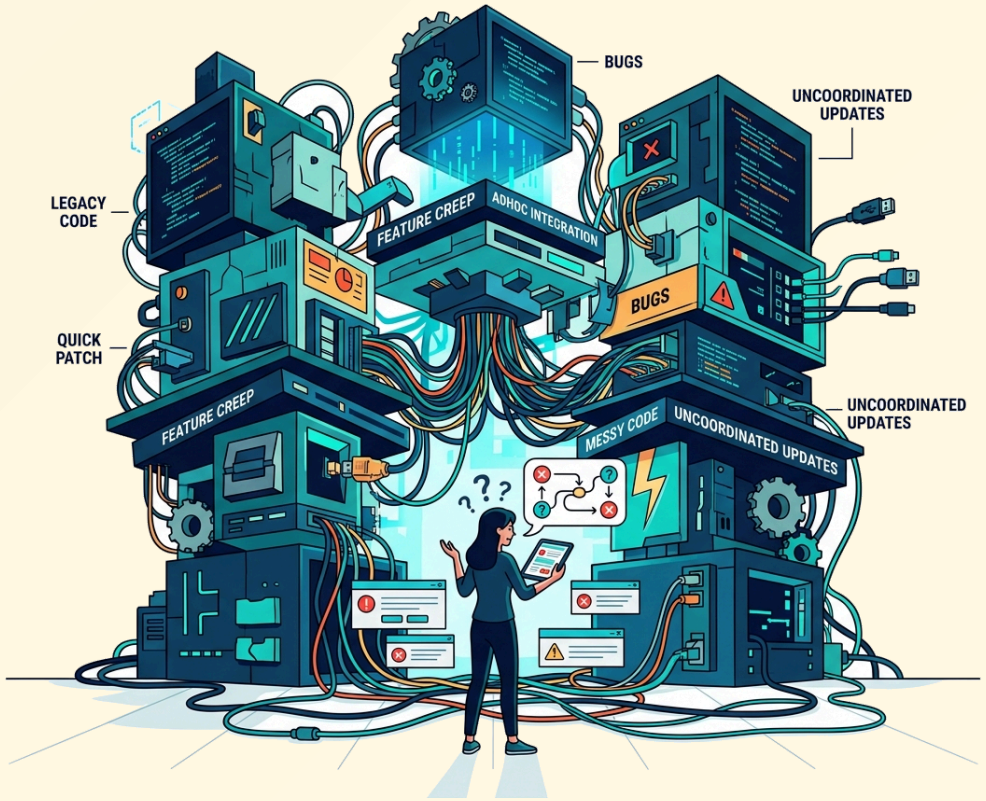
Visual Comparison: UI vs. UX in Real-World Products





Why UX Engineering Matters in Software Engineering

- **1. The Economic ROI (Boehm's Law in UX):**
  - Fixing a workflow flaw during **UX Prototyping** costs **$1** (5 minutes in a wireframe).
  - Fixing it after **Backend API & UI Coding** costs **$10–$20**; fixing it after **Production Release** costs **$50–$100+** in rework and churn.
- **2. Operational Throughput & Error Reduction:**
  - Every extra click or ambiguous label multiplied across 50,000 daily courier dispatches wastes thousands of labor hours.
- **3. Safety-Critical UX Disasters:**
  - **Therac-25 (1985)**: Cryptic codes (`Malfunction 54`) and mode confusion caused lethal overdoses.
  - **Hawaii Missile False Alarm (2018)**: Ambiguous text links with no confirmation dialog triggered statewide panic.



Structural Frameworks: Garrett's 5 Planes & Morville's Honeycomb

> ***Architectural View of UX:*** *UX is built from the* ***bottom up****—from abstract business/user strategy to concrete visual pixels.*

1. Jesse James Garrett's 5 Planes of UX (Bottom → Top)

1. **Surface Plane (Concrete)**: Visual sensory design—color palettes, typography, icons, and micro-animations.
2. **Skeleton Plane**: Interface layout, wireframes, navigation bars, and information presentation placement.
3. **Structure Plane**: **Interaction Design (IxD)** (system behavior over time) and **Information Architecture (IA)** (taxonomy).
4. **Scope Plane**: Functional specifications (feature set) and content requirements (*aligns with SRS!*).
5. **Strategy Plane (Abstract)**: User needs (Personas/JTBD) + Business objectives (ROI/KPIs).

2. Peter Morville's 7 Facets (The UX Honeycomb)

- **Useful**: Does the system solve a real, meaningful problem?
- **Usable**: Is the interface easy and intuitive to operate?
- **Findable**: Can users locate features and content when needed?
- **Credible**: Do users trust the system's security and accuracy?
- **Accessible**: Can users with diverse abilities (visual, motor, auditory, cognitive) use it effectively?
- **Desirable**: Does the aesthetic and brand evoke positive emotion?
- **Valuable**: Does it deliver measurable value to users and sponsors?

Conceptual Architecture: Garrett's 5 Planes & The UX Honeycomb

Structural Frameworks of UX: Garrett's 5 Planes & Morville's UX Honeycomb

Engineering User Experience from Abstract Strategy to Concrete Pixels (Left) & Evaluating the 7 Facets of UX Quality (Right)

1. JESSE JAMES GARRETT'S 5 PLANES OF UX (Bottom-Up Architecture)

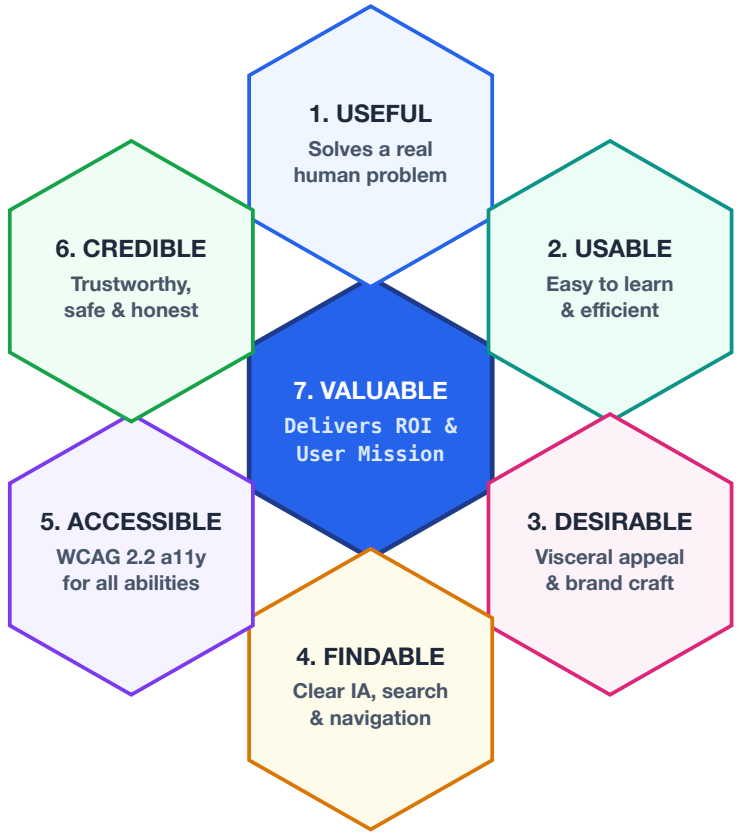
Each plane depends on decisions locked in the plane directly beneath it (Abstract → Concrete)

	Product as Functionality (Software)	Product as Information (Content)
ABSTRACT ——▲—— CONCRETE		
5. SURFACE (Concrete UI)	Sensory & Visual Design	
	Typography, WCAG color contrast, iconography, elevation shadows & micro-anim	
	Deliverable: Design Tokens, High-Fidelity Figma Specs & CSS	
4. SKELETON (Wireframes)	Interface Design & Navigation Design	Information Design
	Button placement (Fitts's Law), form controls, global/local navbars & Gestalt visu	
	Deliverable: Mid-Fidelity Wireframes (Empty, Loading, Ideal, Err	
3. STRUCTURE (Flows & IA)	Interaction Design (IxD)	Information Architecture (IA)
	State transition flows, error prevention & recovery	LATCH taxonomy, Card Sortin
	Deliverable: Sitemaps, User Flows & UML Activity / State Diagram	
2. SCOPE (SRS Specs)	Functional Specifications	Content Requirements
	Feature set boundaries, User Stories, Acceptance Criteria & Non-Functional Usab	
	Deliverable: Software Requirements Specification (SRS) & Product	
1. STRATEGY (Foundation)	User Needs (Personas & JTBD)	Business Objectives
	Empirical field research, user goals & pain points	Conversion ROI, retention KPIs
	Deliverable: Evidence-Based Personas, JTBD & Product Vision	

[Architectural Rule]: Designing Surface UI (Plane 5) without Strategy/Scope/Structure causes project failure!

2. PETER MORVILLE'S 7 FACETS OF UX (The UX Honeycomb)

Moves beyond pure "Usability" to evaluate holistic product value across 7 dimensions



Engineering Synthesis: A system with 99.99% uptime that isn't Useful, Findable, or Accessible fails its users.

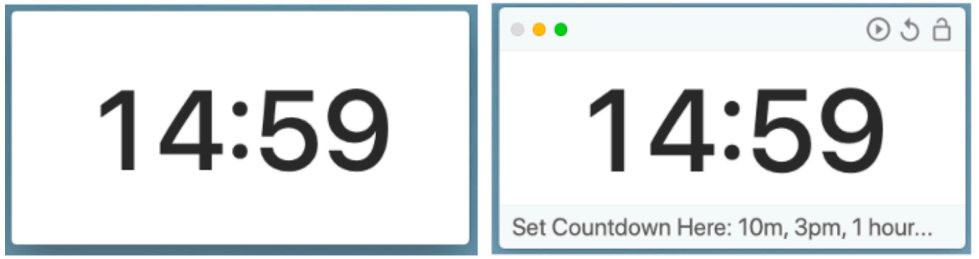
ISO 9241-11 & Jakob Nielsen's 5 Quality Components of Usability

> ***ISO 9241-11 Definition:*** *Usability is the extent to which a system can be used by* ***specified users*** *to achieve* ***specified goals*** *with* ***effectiveness, efficiency, and satisfaction*** *in a* ***specified context of use****.*

Nielsen's 5 Usability Dimensions	Core Engineering Question	Empirical Measurement Metric (Quantitative)
1. Learnability	How easily can a **first-time novice** accomplish basic tasks upon first encounter?	First-attempt **Task Completion Rate (%)** & time to complete onboarding without assistance.
2. Efficiency	Once users have learned the system, **how quickly** can they perform frequent tasks?	**Time-on-Task (seconds)**, number of clicks/keystrokes, or orders processed per minute.
3. Memorability	After returning from a hiatus (weeks/months), how easily can users **re-establish proficiency**?	Task completion time & error count after a **30-day non-use interval**.
4. Errors	How many errors do users make, how **severe** are they, and how easily can they **recover**?	**Error Rate per Task**, frequency of critical dead-ends, and mean time to recover.
5. Satisfaction	How pleasant, comfortable, and frustration-free is the subjective experience?	Post-study standardized scales: **SUS (0–100)**, **CSAT**, or **Net Promoter Score (NPS)**.

The Aesthetic-Usability Effect & Norman's Emotional Design

- **1. The Aesthetic-Usability Effect (Kurosu & Kashimura, 1995):**
  - Tested 26 ATM layouts with identical buttons and logic.
  - Visually attractive layouts were rated **significantly easier to use** (`r = 0.589`), fostering trust and patience for minor hiccups —though aesthetics **never** rescue severe task failures!
- **2. Norman's 3 Levels of Emotional Design:**
  - **Visceral (`< 50ms`)**: Pre-conscious gut reaction to visual symmetry, color harmony, and craft.
  - **Behavioral (In-Use)**: Subconscious feel of functional usability, speed, and control.
  - **Reflective (Post-Use)**: Conscious pride, brand meaning, and peer recommendation.



一開始的呈現是很簡約的，使用者點擊後出現
其他的選單。同時滿足簡約與彈性。

Concept Check Question 1 (CCQ 1)

A hospital deploys a new Emergency Room Triage System. First-time nursing interns complete training in 10 minutes (**high novice learnability**), but veteran ER nurses require **14 mouse clicks across 5 modal dialogs** to register a single critical patient during mass-casualty surges. According to **Jakob Nielsen's 5 Quality Components of Usability**, which specific usability dimension is severely compromised?

- **A. Memorability**, because veteran nurses must re-learn the triage taxonomy after returning from annual leave.
- **B. Efficiency**, because expert users face excessive interaction steps when executing high-frequency tasks.
- **C. Learnability**, because first-time nursing interns cannot complete basic triage registration without manuals.
- **D. Visceral Aesthetics**, because the modal dialogs lack modern gradient styling and smooth micro-animations.

6.2 Cognitive Psychology & Human Information Processing

> *"People are not machines. When a user makes an error, it is almost always a design error—not a human error."*
> *— Don Norman (The Design of Everyday Things)*

Pioneers of UX & Cognitive Engineering: Don Norman

- **Role & Distinctions:**
  - Distinguished Professor of Cognitive Science, UC San Diego; Former VP of Advanced Technology, **Apple**; Co-founder, **Nielsen Norman Group (NN/g)**.
- **Originator of the Term "*User Experience (UX)*" (1993):**
  - Created the world's first "*User Experience Architect*" title at Apple to encompass the entire human experience of a product beyond the screen UI.
- **Seminal Contributions to HCI & Software Engineering:**
  - Authored ***The Design of Everyday Things*** (originally *The Psychology of Everyday Things*, 1988).
  - Formulated the **7 Stages of Action**, the **Gulfs of Execution & Evaluation**, **Perceived Affordances & Signifiers**, and the **Triad of Mental Models**.
- **Signature Engineering Maxim:**

> *"Do not blame the user! Human error is usually the result of poor design that ignores cognitive psychology."*



Don Norman
UC San Diego / Apple / NN/g Co-Founder

Norman's 7 Stages of Action, The Two Gulfs & MHP Thresholds

> ***Core Interaction Cycle (Don Norman & Card/Moran/Newell):*** *Every human interaction bridges* ***Internal Psychological Goals*** *and* ***External Physical System State*** *within biological clock limits (`100ms / 1s / 10s`).*

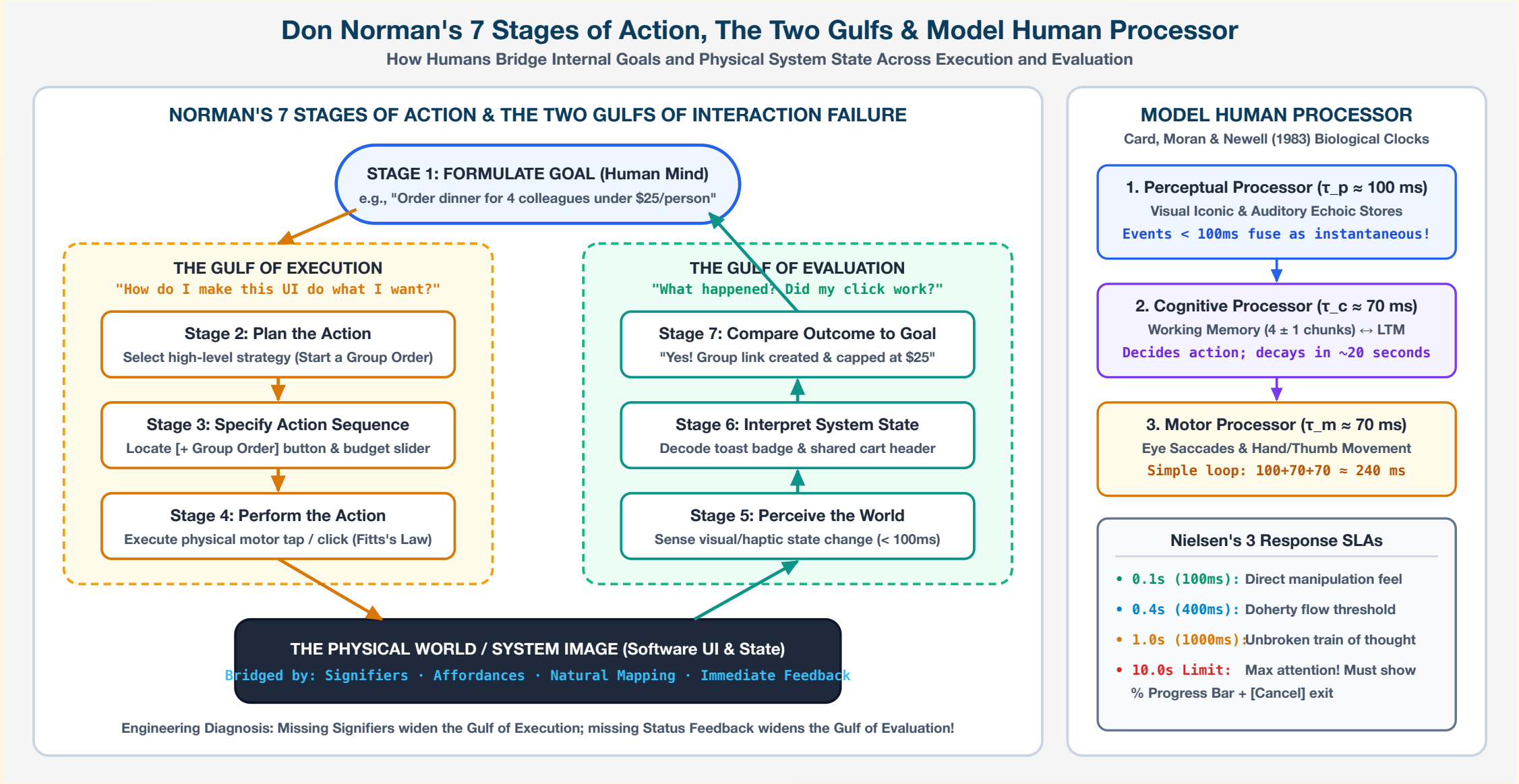
1. The 7 Stages & The Two Gulfs

- **Goal (Stage 1)**: Formulate the high-level objective ("*Order dinner for $15*").
- **The Gulf of Execution ("*How do I work this?*")**:
  - **2. Plan** → **3. Specify** → **4. Perform** physical action.
  - *Bridged by*: Clear **Signifiers**, **Affordances**, **Natural Mapping**, and **Constraints**.
- **The Gulf of Evaluation ("*What happened? Did it work?*")**:
  - **5. Perceive** → **6. Interpret** → **7. Compare** to Goal.
  - *Bridged by*: Immediate **Status Feedback (`< 100ms`)** and clear **Conceptual Models**.

2. Model Human Processor (MHP) & Latency SLAs

- **Biological Cycle Times (Card, Moran & Newell, 1983)**:
  - **Perceptual (`τ_p ≈ 100 ms`)** + **Cognitive (`τ_c ≈ 70 ms`)** + **Motor (`τ_m ≈ 70 ms`)** = `~240 ms` simple reaction loop.
- **Jakob Nielsen's 3 Universal Response-Time Limits**:
  - `0.1s (100 ms)`: Feels **instantaneous** (direct manipulation: button ripple, scroll tracking).
  - `1.0s (1,000 ms)`: User's **flow of thought stays unbroken** without needing a progress bar.
  - `10.0s (10,000 ms)`: Max attention limit! Must show a `%` **Progress Bar** + time estimate + `[Cancel]` **exit**.

Conceptual Model: 7 Stages of Action, The Two Gulfs & MHP



Affordances, Signifiers, Mappings & Feedback

> ***Bridging the Gulfs:*** *Don Norman refined his framework to distinguish* ***Affordances*** *(what actions are possible) from* ***Signifiers*** *(the perceptible clues showing where and how to act).* ”

Core Design Concept	Formal Definition in HCI / UX	Software UI Engineering Application
1. Affordance	The relationship between a physical/digital object and a user that defines **what actions are possible**.	A scrollable viewport *affords* vertical panning; a text input box *affords* typing characters.
2. Signifier	Any **perceptible visual, auditory, or tactile clue** that communicates *where* and *how* an action should be taken.	An underlined blue label, a raised button shadow, a `[Search]` icon, or a visible scrollbar thumb.
3. Natural Mapping	The spatial or logical relationship between **controls and their effects** in the world.	Pressing the **Right Arrow** `→` advances to the next slide; dragging a slider up increases volume.
4. Feedback	Communicating the **results of an action** back to the user immediately and clearly.	Button ripple (`< 100ms`), toast confirmation (`"Added to Cart"`), or inline red validation text.
5. Constraints	Limiting possible actions (Physical, Cultural, Semantic, Logical) to **prevent invalid states**.	Disabling the `"Place Order"` button until a valid delivery address and payment method are selected.

Case Study: Norman's Door & Signifier Failure



Mental Models vs. Conceptual Models & Cognitive Load Theory

> ***Norman's Triad & Sweller's Cognitive Load (1988):*** *Designers communicate with users* ***only through the System Image****, while human* ***Working Memory*** *holds only* `4 ± 1` ***active chunks*** *(* `~20s` *decay).*

1. Norman's Triad of Models

- **1. Designer's Conceptual Model (Implementation)**:
  - Normalized SQL tables, foreign keys, Kafka partitions, and state machines.
- **2. User's Mental Model (Everyday Beliefs)**:
  - Shopping carts, paper receipts, folders, and instant Undo.
- **3. The System Image (The Bridge)**:
  - The visible UI, labels, metaphors, and error messages.
  - **[Anti-Pattern] Implementation Model Leak**: Exposing raw SQL errors (`ORA-02292`) or internal shard IDs to end users breaks their Mental Model!

2. Sweller's 3 Types of Cognitive Load

- **Intrinsic Load (Inherent Task Complexity)**:
  - Managed by breaking complex flows into **step-by-step wizards**.
- **Extraneous Load (Bad UI Friction)**:
  - Wasted mental effort caused by visual clutter, inconsistent menus, or forcing users to memorize promo codes across screens. **Goal: Reduce to ~0%!**
- **Germane Load (Schema Mastery)**:
  - Productive effort invested in learning reusable patterns.

Conceptual Diagram: Mental Models & Cognitive Load Architecture

Norman's Triad of Models & Sweller's Cognitive Load Architecture

Aligning the System Image with the User's Mental Model (Left) & Eliminating Extraneous Working Memory Load (Right)

1. NORMAN'S TRIAD: DESIGNER MODEL vs. USER MENTAL MODEL

Designers NEVER communicate directly with users—only through the System Image!

DESIGNER'S MODEL

(Implementation View)

- Normalized SQL Foreign Keys
- Kafka Outbox Partition Shards
- Git SHA-1 Merkle DAG Blobs
- REST JSON Status Codes

NO DIRECT TELEPATHY!

USER'S MENTAL MODEL

(Everyday Domain View)

- Shopping Cart & Receipt
- Kitchen Preparing My Meal
- Folders & Trash Can on Desk
- Instant Undo When I Slip

Encodes

Perceives & Acts

THE SYSTEM IMAGE (The Only Bridge)

Visible UI + Affordances + Labels + Feedback + Errors

[Good UX]: Translates DB state into human concepts ("Card Expired")

[Implementation Leak]: Exposes "SQLSTATE[23505] idx_users_uq"

Rule: Design the System Image to match the User's Mental Model!

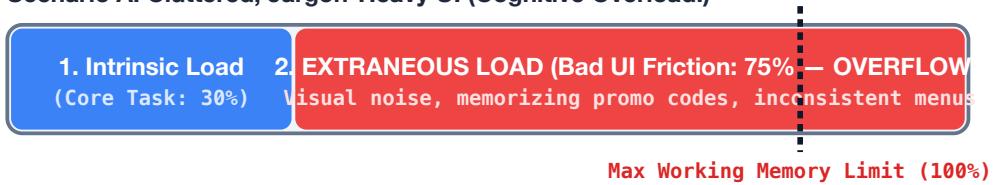
Why "Implementation Leaks" Cause Severe User Errors:

When engineers expose backend tables, flags, or RPC timeouts directly in the UI, users form broken Mental Models and make catastrophic Mistakes (Heuristic #2 & #5).

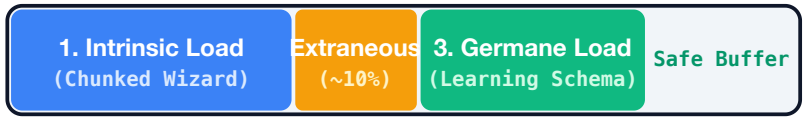
2. SWELLER'S COGNITIVE LOAD & WORKING MEMORY LIMITS

Human Working Memory holds only 4 ± 1 active chunks (Cowan, 2001) for ~20 seconds

Scenario A: Cluttered, Jargon-Heavy UI (Cognitive Overload!)



Scenario B: Well-Engineered UX (Extraneous Load Minimized!)



3 UX Engineering Tactics to Slash Cognitive Load

1. **Recognition Over Recall (Heuristic #6):**
   Display clickable recent addresses, dish photos & one-tap coupons instead of typing codes from mem
2. **Progressive Disclosure & Chunking (Hick's Law):**
   Break 16-field registration into a 3-step progress bar (4 fields per screen) matching the 4 ± 1 chunk lim
3. **Defeat Change Blindness (Foveal Attention ~2° Visual Angle):**
   Place inline validation errors & spinners right next to the clicked button—never in a distant corner!

Concept Check Question 2 (CCQ 2)

In a cloud deployment dashboard, an engineer clicks the **"Deploy to Production"** button. Because the frontend lacks a loading spinner, toast message, or state change for 8 seconds, the engineer assumes the click failed and clicks the button three more times, triggering conflicting deployments. In **Don Norman's Action Cycle**, which specific breakdown caused this error?

- **A.** A wide **Gulf of Execution**, because the engineer could not locate the deployment button on the screen.
- **B.** Excess **Intrinsic Cognitive Load**, because cloud deployment inherently requires memorizing Linux kernel flags.
- **C.** A wide **Gulf of Evaluation**, because the interface failed to provide immediate feedback on the system's new state.
- **D.** A **Natural Mapping** violation, because the deployment button was positioned on the right side of the toolbar.

6.3 Quantitative Laws of Interaction & Visual Perception

> *"Design is a science as much as an art: human motor movement, choice reaction time, and visual grouping obey mathematical laws."*

Pioneers of Interaction Design: Ben Shneiderman

- **Role & Distinctions:**
  - Distinguished University Professor of Computer Science, University of Maryland; Founding Director, **Human-Computer Interaction Lab (HCIL)**; IEEE & ACM Fellow; NAE Member.
- **Pioneered Concepts & Inventions:**
  - Coined **Direct Manipulation (1982)**: Continuous visual representation of objects, physical actions/pointing instead of complex command syntax, and rapid reversible incremental operations.
  - Invented the **highlighted clickable web hyperlink**, dynamic sliders, **Treemaps** for hierarchical data visualization, and **Human-Centered AI (HCAI)**.
- **Author of the Classic Textbook:**
  - *Designing the User Interface* (**Eight Golden Rules of Interface Design**: Consistency, Universal Usability, Informative Feedback, Closure, Error Prevention, Easy Reversal, Internal Locus of Control, Short-Term Memory Load).



Ben Shneiderman
Univ. of Maryland HCIL / ACM Fellow

Mathematical & Behavioral Laws: Fitts, Hick, Jakob & Tesler

> ***Predictive HCI Modeling:*** *Human motor targeting (`Fitts`), decision latency (`Hick`), convention transfer (`Jakob`), and complexity conservation (`Tesler`) follow strict engineering laws.*

1. Fitts's Law & Hick-Hyman Law

- **Fitts's Law (1954 — Motor Targeting Speed)**:
  - `MT = a + b · log2(D / W + 1)` (`D` = Distance, `W` = Target Width).
  - **UI Rules**: Make primary CTAs large (`>= 44×44 px` / `48×48 dp`, `↑ W`) inside the mobile **Bottom Thumb Zone (`↓ D`)**; pin global menus to **Screen Edges/Corners (`W = ∞`)**; isolate destructive buttons!
- **Hick-Hyman Law (1952 — Choice Reaction Time)**:
  - `RT = a + b · log2(n + 1)` (`n` = number of choices).
  - **UI Rules**: Apply **Progressive Disclosure** (show 3–5 core choices + `[Recommended]` default; tuck rare options under `"More..."`).

2. Jakob's, Tesler's, Postel's & Doherty Laws

- **Jakob's Law**: Users spend most of their time on *other* sites—follow standard conventions (top-right `[Cart]`, `Ctrl+Z`) so navigation takes **0% learning load**.
- **Doherty Threshold (`< 400 ms`)**: Keep system-human interaction under `400 ms` via **Skeleton Screens & Optimistic UI**.
- **Tesler's Law (Conservation of Complexity)**: Shift irreducible complexity from the User to the Code (e.g., auto-fill City/State from ZIP code).
- **Postel's Law (Robustness Principle)**: Accept flexible human input (`0912-345-678` or `0912345678`) liberally and normalize in code!

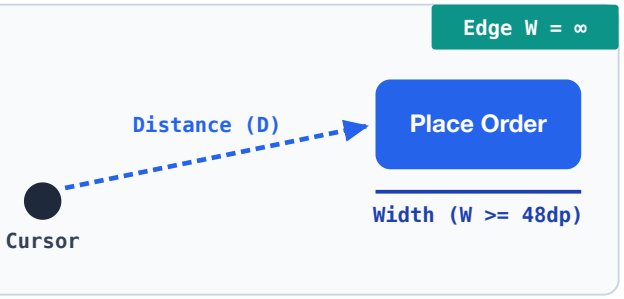
Conceptual Diagram: Fitts's Law, Hick's Law & Gestalt Grouping

Quantitative Laws of Interaction (Fitts & Hick) & Gestalt Visual Perception

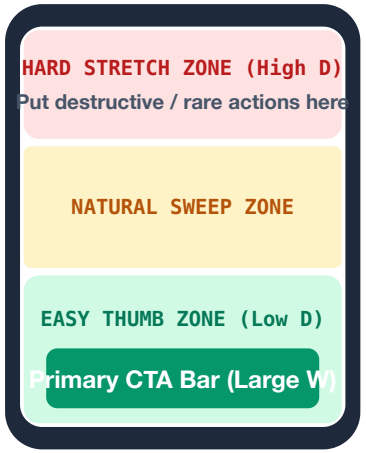
Mathematical Optimization of Pointing Speed (Left), Choice Reaction Time (Center) & Perceptual Grouping (Right)

1. FITTS'S LAW (Motor Pointing)

$$MT = a + b \cdot \log_2(D / W + 1)$$

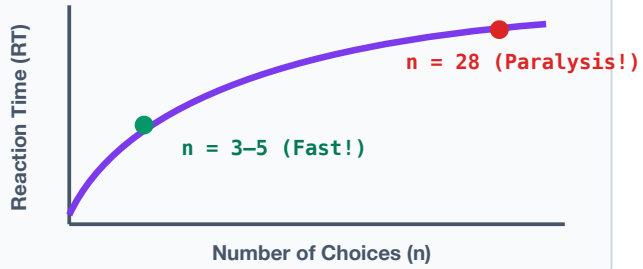


Mobile One-Handed Thumb Ergonomics



2. HICK-HYMAN LAW (Decision Time)

$$RT = a + b \cdot \log_2(n + 1)$$



Solution: Progressive Disclosure & Smart Defaults

[Anti-Pattern] n=28

- ○ Bank 01 ○ Bank 15
- ○ Bank 02 ○ Bank 16
- ○ Bank 03 ○ Wallet A
- ○ Bank 04 ○ Wallet B
- ○ Bank 05 ○ Wallet C
- ○ Bank 06 ○ Voucher
- ○ Bank 07 ○ Crypto

Linear Scan Overload!

[Best Practice] n=3

- • Apple Pay [Default] 1-Tap Biometric
- ○ Saved Visa ••4210
- ▸ More Methods (26)...

Decision Time < 1.2s!

3. GESTALT VISUAL GROUPING LAWS

How the visual cortex groups UI controls without clutter

1. Law of Proximity (Whitespace Grouping)

Delivery Address (4px gap to input) | Courier Note (20px separation)

Closer items = Same group (No heavy boxes needed!)

2. Law of Similarity (Visual Consistency)

Add + | Pay Now | Cancel

Same color/shape = Same interactive behavior

3. Common Region (Cards) & 4. Closure

Card 1 (Enclosed) | Card 2 (Enclosed) | C3..

Clipped 3rd card triggers Closure → "Scroll right!"

5. Figure-Ground & WCAG 2.2 Contrast

- Modal backdrop dimming separates Figure from Ground
- WCAG Level AA requires >= 4.5 : 1 text contrast
- • Never use Red/Green color alone: Add [!] Icon + Text!

Gestalt Visual Hierarchy & WCAG 2.2 Accessibility Engineering

- **1. Visual Hierarchy & Eye-Tracking (F / Z Patterns):**
  - Users **scan** interfaces rather than reading word-for-word.
  - Enforce a clear typographic scale (`H1 > H2 > Body`), line width `50-75 chars` (`65ch`), and **Gestalt Proximity** whitespace.
- **2. W3C WCAG 2.2 Accessibility (POUR Mandate):**
  - **Luminance Contrast (Level AA)**: Minimum `4.5 : 1` for normal text (`3.0 : 1` for large text `>= 18pt` and UI controls).
  - **Color-Blindness Safety**: Never rely on Red vs. Green alone (~8% of males have Deuteranopia/Protanopia)—always pair color with `[!] Icon + Text Label`.
  - **Keyboard & Screen Reader**: Semantic `<button>`, focus rings, and `aria-label`.



https://www.mockplus.com/blog/post/bad-web-design

Peer Discussion 6.3: Diagnosing a High-Friction Checkout Screen

UX Engineering Audit: Refactoring the Food Delivery Checkout UI (4 Mins)

- **Scenario:** A junior developer built the checkout screen for our **Food Delivery App**, and cart abandonment jumped to **62%**. An inspection reveals four design flaws:
  1. The **"Place Order"** button is a tiny `24×18 px` text link at the top-left corner of the mobile screen, right next to **"Empty Entire Cart"** (`2px` gap).
  2. The payment selector displays all **28 supported banks and wallets** in a single unsorted radio-button list.
  3. The phone number field rejects `0912-345-678` with a red popup: "*Error: Numeric digits only! Re-enter all 8 address fields.*"
  4. Required fields are marked *only* by changing their border color from green to red (`#ef4444` on `#e5e7eb` gray background, contrast `2.1:1`).
- **Discussion Tasks:**
  - **Q1.** Identify which specific **Law of UX or Accessibility Standard** (Fitts's, Hick's, Tesler's/Postel's, WCAG) is violated by each of the 4 flaws (`#1` to `#4`).
  - **Q2.** Propose the concrete **UI refactoring fix** for each flaw!



Concept Check Question 3 (CCQ 3)

According to the Shannon formulation of **Fitts's Law** (`MT = a + b · log2(D / W + 1)`), why do desktop operating systems place critical global menus (such as the macOS Menu Bar or the Windows Start button) flush against the **extreme edges or corners of the screen**?

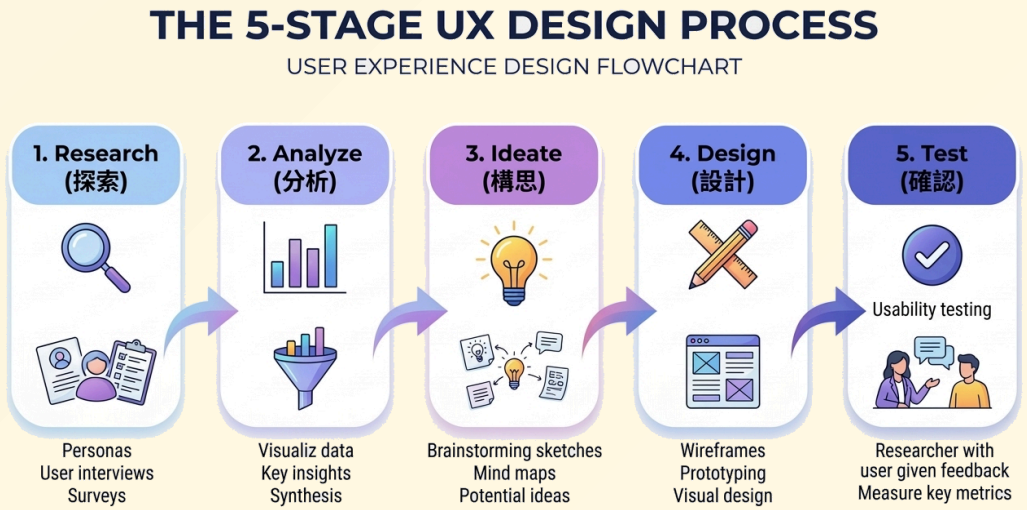
- **A.** Screen edges reduce the number of menu items `n`, lowering cognitive choice reaction time under Hick's Law.
- **B.** The monitor bezel blocks the cursor from overshooting, giving edge targets an effectively infinite width `W`.
- **C.** Placing menus at the screen edge increases luminance contrast ratios to satisfy WCAG 2.2 Level AAA standards.
- **D.** Peripheral vision processes corner pixels in `< 20ms`, bypassing the Model Human Processor's motor stage.

6.4 The Human-Centered Design (HCD) Process & User Research

> *"You are not the user! Every assumption you make about how people will use your software is a hypothesis waiting to be tested."*
> *— Jakob Nielsen & Don Norman*

The Double Diamond Lifecycle & ISO 9241-210 HCD Process

- **1. Diamond 1 — Problem Space ("*Design the Right Thing*"):**
  - **Discover (Diverge <)**: Field observation (**Contextual Inquiry**) and user interviews to uncover real pain points.
  - **Define (Converge >)**: Synthesize findings into **Personas**, **JTBD**, and a focused Problem Statement (**SRS Scope**).
- **2. Diamond 2 — Solution Space ("*Design the Thing Right*"):**
  - **Develop (Diverge <)**: Explore Information Architecture, low-fidelity sketches, and wireframes.
  - **Deliver (Converge >)**: Run iterative usability tests, refine UI tokens, and ship in agile sprints.



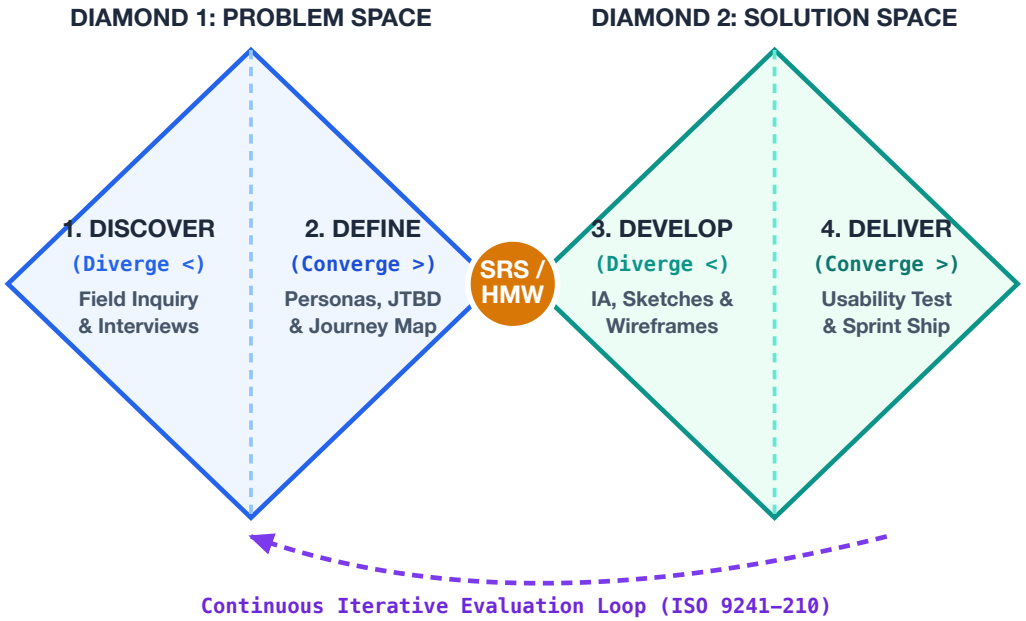
Conceptual Diagram: Double Diamond & 2×2 User Research Matrix

The Double Diamond HCD Lifecycle & Empirical User Research Landscape

Diverging and Converging Across Problem & Solution Spaces (Left) + Triangulating Qualitative & Quantitative Methods (Right)

1. UK DESIGN COUNCIL DOUBLE DIAMOND & ISO 9241-210

"Design the Right Thing" (Diamond 1) Before "Designing the Thing Right" (Diamond 2)



Direct Traceability from UX Artifacts to Software Engineering (Ch 03–05)

- Evidence-Based Persona ——▶ UML Actor + Non-Functional Usability SLAs (e.g., 56dp outdoor targets)
- Jobs-to-be-Done (JTBD) ——▶ Agile Epics & User Stories ("When [Context], I want [Action], so I can...")
- Hierarchical Task Analysis (HTA) ——▶ UML Use Case Scenarios, Activity Diagrams & BDD Acceptance Tests

2. THE 2×2 USER RESEARCH LANDSCAPE (NN/g)

Triangulate What Users Do (Behavioral) vs. What Users Say (Attitudinal)

	QUALITATIVE ("Why?")	QUANTITATIVE ("How Much?")
BEHAVIORAL (What They Do)	**Field & Lab Observation** • Contextual Inquiry (Field) • Moderated Usability Test (Think-Aloud Protocol) • Open/Closed Card Sorting `Sample: N = 5–12 users` Finds mental-model bugs!	**Scale Telemetry & RCTs** • Clickstream Funnel Logs • Controlled A/B Testing • Tree Testing (First-Click) • Eye-Tracking Heatmaps `Sample: N = 100–100k+` Pinpoints drop-off rates!
ATTITUDINAL (What They Say)	**Deep Needs & Motivations** • Semi-Structured Interviews • Longitudinal Diary Studies • Stakeholder Interviews • Focus Groups `Sample: N = 8–15 users` Discovers JTBD & Context	**Standardized Scales** • SUS (0–100 Usability Scale) • SEQ & NASA-TLX Workload • Google HEART Happiness • Kano Feature Priority Survey `Sample: N = 30–1,000+` Benchmarks satisfaction!

Golden Rule: Never rely on Focus Groups alone—observe real behavior!

Synthesizing Research: Personas, JTBD, Journey Maps & HTA

> ***From Raw Field Notes to Engineering Models:*** *Distill qualitative observations into* ***Goal-Directed Personas*** *(Who),* ***Jobs-to-be-Done*** *(Why),* ***Customer Journey Maps*** *(Macro Timeline), and* ***Hierarchical Task Analysis*** *(Micro Steps).*

1. Personas (Alan Cooper) & JTBD (Christensen)

- **Alan Cooper's Goal-Directed Personas (*The Inmates Are Running the Asylum*)**:
  - Empirical behavioral archetypes clustered via **Affinity Diagramming (KJ Method)**—never fictional marketing stereotypes!
  - Captures **Context of Use** (e.g., "*Courier Wei, operates one-handed on a scooter in rain*"), **Goals**, and **Frustrations**.
- **Jobs-to-be-Done (JTBD Template)**:
  - **When** `[Situation]`, **I want to** `[Action]`, **so I can** `[Outcome]`.
  - *Example*: "*When working overtime with 4 colleagues at 8 PM, I want to share a group cart link capped at $25/person, so we can order dinner without passing my phone around.*"

2. Customer Journey Maps & Task Analysis (HTA)

- **Customer Journey Map (Macro Experience & Emotion Curve)**:
  - Plots user actions, thoughts, and **Emotional Peaks/Valleys** (`Delight (+) ↔ Frustration (−)`) across timeline stages.
  - Exposes the infamous **"Post-Payment Anxiety Valley"** in food delivery!
- **Hierarchical Task Analysis (HTA — Micro Task Tree)**:
  - Decomposes `Goal 0` into numbered sub-tasks (`1, 2, 3...`) and conditional execution `Plans`—mapping directly to **UML Activity Diagrams** and **BDD Acceptance Tests**!

Visual Case Study: Customer Journey Map in Food Delivery

Customer Journey Map & Emotional Curve: Food Delivery Group Order

Mapping Touchpoints, User Thoughts, Emotional Valleys & Software Architecture Opportunities Across Time

	Stage 1: Group Setup	Stage 2: Shared Cart	Stage 3: Checkout & Pay	Stage 4: Kitchen Wait	Stage 5: Live Arrival
1. User Actions	• Picks restaurant at 7:45 PM • Taps [+ Group Order] • Shares deep link on Slack	• 3 coworkers add dumplings • 1 coworker walks away • Host checks who is ready	• Applies promo code • Reviews $25/person cap • Pays via Apple Pay	• Waits 25 mins hungry • Checks phone 6 times • "Did the store get it?!"	• Watches rider on GPS map • Meets rider at lobby • 1-tap auto-split receipt
2. Emotion Curve (+) Delight (−) Anxiety	Excited (+): Fast 1-Tap Share	Friction (-): 1 Member Stalls Cart	Relieved (+): Biometric Pay	ANXIETY VALLEY (Heuristic #1 Gap!) "Black-Box Post-Payment Silence"	Peak Delight (+): Live Map
3. SE & UX Solutions	**IA & First-Click Fix** • Place [+ Group Order] in Restaurant Menu Header • Auto-generate QR + Web deep link (No app install required for guests!)	**Edge-State Wireframe** • Show real-time badges: `[ Picking... ] vs [ Ready ✓ ]` • Add Countdown Timer + [Lock Cart & Checkout] host override button!	**Hick's & Tesler's Laws** • Pre-select best coupon automatically in code • Default to last-used 1-tap payment method • Upfront fee transparency	**Healing the Anxiety Valley** • Push WebSocket events: `Confirmed → Wok → Packed` • Show dynamic ETA range • Provide 60s Undo/Cancel window (Heuristic #1 & #3)	**Peak-End Rule Delight** • Stream Redis GeoHash rider coordinates (<5ms) • Photo proof at drop-off • Auto-email itemized bill to each group member

6.5 Information Architecture, Wireframing & Prototyping

> *"If a user can't find it, it doesn't exist."*
> *— Steve Krug (Author of Don't Make Me Think)*

Information Architecture (IA), Card Sorting & Atomic Design

> ***Structure Before Pixels:*** *Never let backend database schemas dictate your UI menu hierarchy—validate mental models via* ***Card Sorting & Tree Testing****, and scale UI via* ***Brad Frost's Atomic Design****.*

1. IA Systems, Card Sorting & Tree Testing

- **4 IA Systems (Rosenfeld & Morville)**:
  - **Organization (LATCH)**: **L**ocation, **A**lphabet, **T**ime, **C**ategory, **H**ierarchy.
  - **Labeling, Navigation (`Global / Local / Breadcrumb`) & Search**.
- **Empirical IA Validation**:
  1. **Card Sorting (Generative — Bottom-Up)**: `15-20` users group feature cards into piles (**Open Sort** invents category labels; **Closed Sort** tests fixed buckets).
  2. **Tree Testing (Evaluative — Reverse Card Sort)**: Tests a plain-text menu tree for **Findability Success (%)** and **First-Click Directness (%)** (`87%` win if 1st click is right vs. `46%` if wrong!).

2. Brad Frost's Atomic Design System (2013)

- **Single Source of Truth (`Figma ↔ React / iOS / Android`)**:
  - **0. Design Tokens**: Semantic JSON variables (`--color-primary`, `--space-md`) enforcing **Dark Mode & WCAG contrast**.
  - **1. Atoms**: Indivisible primitives (`<Button>`, `<Input>`, `<Icon>`).
  - **2. Molecules**: Simple functional groups (`<PromoCodeBox>`).
  - **3. Organisms**: Self-contained UI sections (`<RestaurantCard>`).
  - **4. Templates**: Content-agnostic page layout grids.
  - **5. Pages**: Real dynamic payloads testing edge cases (`80-char` dish names).

Conceptual Diagram: Card Sorting, Tree Testing & Atomic Design

Information Architecture Validation & Brad Frost's Atomic Design System

Empirical Taxonomy Discovery via Card Sorting & Tree Testing (Left) + Component-Driven UI Engineering (Right)

1. EMPIRICAL IA: CARD SORTING vs. TREE TESTING

Validate mental-model findability BEFORE drawing visual screens!

Step A: Card Sorting (Generative — Bottom-Up Taxonomy)

"Tip Courier"
"Tax Invoice"
"Missing Drink"

15–20 Users →

Pile: "My Orders"
- Tip Courier
- Missing Drink
- Live Map

Pile: "Billing"
- Tax Invoice
- Saved Cards
- FoodPass Plan

`Open Sort = Users invent pile names | Closed Sort = Fixed buckets`

Step B: Tree Testing (Evaluative — "Reverse Card Sorting")

Task Prompt: "Where would you go to report a missing drink?"

- ▸ Home & Restaurants
- ▾ My Orders (First Click ✓)
  - • Active Order Map
  - • Get Help / Missing Item ✓
- ▸ Wallet & Billing
- ▸ Account Settings

Quantitative IA Metrics

1. Success Rate (%): Reached correct destination node?

2. Directness (%): Zero backtracking up the tree?

`First-Click Correct → 87% Win`
`First-Click Wrong → 46% Win`

2. BRAD FROST'S ATOMIC DESIGN SYSTEM (5 Levels + Tokens)

Single Source of Truth connecting Figma Components to Production React / iOS / Android

FOUNDATION: DESIGN TOKENS (JSON → CSS / Swift / Kotlin)

`--color-primary: #2563eb | --space-md: 16px | --radius-card: 10px | WCAG AA Built-In`

1. ATOMS

Indivisible UI primitives

`<Input placeholder />`

`<Button />`

`<Icon /> <Badge />`

2. MOLECULES

Simple functional unit

`<PromoCodeBox />`

Apply

`Input + Button + Label`

3. ORGANISMS

Complete UI section

`<RestaurantCard />`
- Photo + 4.9★ Badge
- ETA 20m + Fee $0
- [+ Add Favorite]

`Reusable across screens`

4. TEMPLATES (Page Skeleton)

Content-agnostic layout wireframe grid

`[ <GlobalNavbarOrganism /> ]`

`[ 2-Column Menu Grid Slot ]`

`[ <StickyCartFooter /> ]`

Defines responsive breakpoints

5. PAGES (Real Data Instance)

Populated with live production payloads

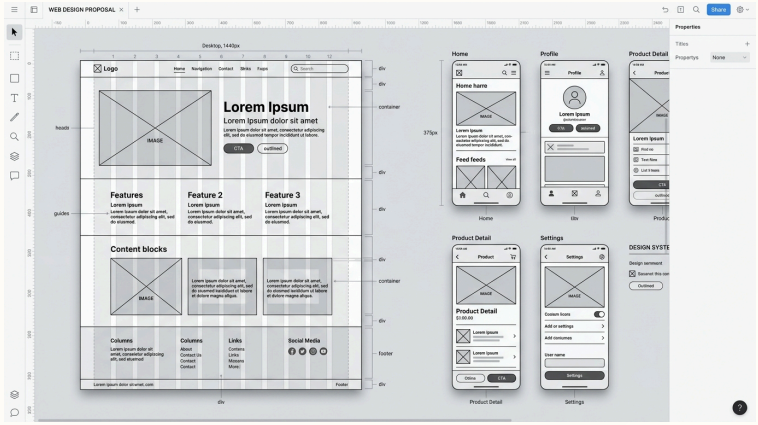
"Din Tai Fung — Group Cart"
- Tests 80-char dish names
- Tests Empty / Error / Loading
- Verifies Heuristic #4 Consistency

`100% Storybook Verified!`

The Prototyping Fidelity Spectrum: Sketch → Wireframe → Prototype

> ***Bill Buxton's Dual Mandate (Sketching User Experiences):*** *Early low-fidelity sketches focus on* ***"Getting the Right Design"*** *(exploring alternatives cheaply); late high-fidelity prototypes focus on* ***"Getting the Design Right"****.*

1. Low & Mid-Fidelity Wireframes (`Cost: Minutes/Hours`)



- **Purpose**: Divergent exploration, IA layout, and locking down the **4 Mandatory UI States** (`Empty` , `Loading` , `Ideal` , `Error`).
- **Sunk-Cost Immunity**: Nobody hesitates to critique or discard a 5-minute grayscale sketch!

2. High-Fidelity Interactive Prototypes (`Cost: Days`)



- **Purpose**: Realistic clickable Figma flows, Design Token contrast verification, summative usability testing, and exact developer handoff.
- **Risk if Used Too Early**: Premature pixel-polishing locks teams into a flawed workflow!

Peer Discussion 6.5: IA & Wireframe States for Group Ordering

UX Architecture Exercise: Designing the "Office Group Order" Flow (4 Mins)

- **Scenario:** You are designing the **Group Order (Shared Cart)** feature for our Food Delivery App based on the JTBD from Section 6.4 (*Host invites 4 colleagues via link; budget cap $25/person*).
- **1. Information Architecture & Labeling:**
  - Where in the app's hierarchy should `"Start Group Order"` live so users find it on their **first click** (Restaurant Menu header vs. Account Settings vs. Checkout Screen)?
- **2. Mid-Fidelity Wireframe & The 4 UI States:**
  - Specify the **Host's Shared Cart Wireframe**, explicitly addressing the **4 Mandatory UI States**:
    - **Empty State**: Host just created the link; 0 colleagues have joined yet. What signifiers/actions are shown?
    - **Loading / Active State**: 3 colleagues have added items; 1 colleague is still "*Choosing...*".
    - **Error / Edge State**: A colleague adds a $32 item exceeding the $25 cap, OR walks away without clicking "*I'm Ready*". How does the UI prevent the Host from getting stuck?



Concept Check Question 4 (CCQ 4)

Before designing any visual screens for a banking portal, a UX engineer presents 30 participants with a **plain-text collapsible menu hierarchy** (stripped of all graphics, colors, and page layouts) and asks: "*Where would you click to freeze a lost debit card?*" Which specific UX evaluation method is being conducted, and what does it measure?

- **A. Open Card Sorting**, which discovers how users invent new category names from unorganized feature cards.
- **B. Tree Testing**, which evaluates menu findability and first-click accuracy within an existing hierarchy.
- **C. A/B Split Testing**, which measures live production conversion rates between two visual button styles.
- **D. Atomic Design Auditing**, which verifies that UI molecules inherit CSS tokens from primitive atoms.

6.6 Analytical Evaluation: Nielsen's 10 Usability Heuristics

> *"Heuristic evaluation is a discount usability engineering method: 3 to 5 independent evaluators can identify 75% to 80% of major usability problems before a single user test is run."*
> *— Jakob Nielsen (1994)*

Pioneers of Usability Engineering: Jakob Nielsen

- **Role & Distinctions:**
  - Co-founder, **Nielsen Norman Group (NN/g)** (with Don Norman); Former Sun Microsystems Distinguished Engineer; Ph.D. in HCI, Technical University of Denmark; **SIGCHI Lifetime Achievement Award**.
- **Champion of "*Discount Usability Engineering*" (1989–1994):**
  - Proved that software teams do not need million-dollar biometric labs to build usable software—combining **Simplified Thinking Aloud (`N = 5` users)**, **Paper Prototyping**, and **Heuristic Evaluation** delivers massive ROI rapidly.
- **Core Contributions to Software Engineering:**
  - Formulated the **10 Usability Heuristics (1994)**, **5 Quality Components of Usability**, **Jakob's Law**, **3 Response-Time Limits (`0.1s/1s/10s`)**, and **F-Pattern Eye-Tracking**.



Jakob Nielsen
Sun Microsystems / NN/g Co-Founder

Heuristic Evaluation: Protocol, Evaluators & Severity Rating

> ***Why Multiple Independent Evaluators? (Nielsen & Landauer, 1993):*** *A single evaluator finds only* `~35%` *of usability problems; aggregating* ***3 to 5 independent evaluators*** *uncovers* `75%–80%` *of issues!*

1. The 4-Step Heuristic Evaluation Protocol

1. **Briefing & Task Scenarios**: Evaluators receive the target Persona, domain context, and core task flows to inspect.
2. **Independent Inspection (Crucial Rule!)**:
   - Each of the **3–5 evaluators** inspects the interface **alone** (Pass 1 for overall flow, Pass 2 auditing each screen against the 10 Heuristics) to prevent groupthink bias!
3. **Aggregation & Debriefing**: Combine duplicate findings into a master defect log, citing the specific **Heuristic (`#1–#10`)** violated.
4. **Severity Rating & Action Plan**: Score each defect (`0–4`) to prioritize engineering sprint fixes.

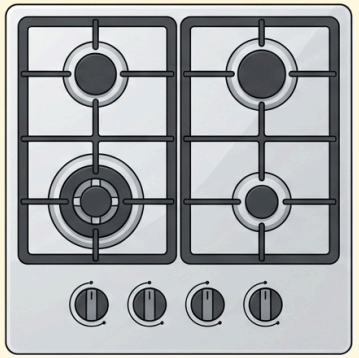
2. Nielsen's 0–4 Usability Severity Scale

- **Severity = f (`Frequency`, `Impact`, `Persistence`)**:
  - `0` — **Not a Problem**: Disagreement; no UX impact.
  - `1` — **Cosmetic Problem**: Fix only if extra time remains in the sprint (e.g., `2px` icon misalignment).
  - `2` — **Minor Usability Problem**: Low priority; causes brief hesitation but users recover quickly.
  - `3` — **Major Usability Problem**: High priority; causes significant confusion, delay, or repeated errors.
  - `4` — **Usability Catastrophe**: **Release Blocker (`P0`)!** Must be fixed before launch (causes data loss, double-charging, or task abandonment).

Nielsen's Heuristics #1 – #3: Status, Real World & User Control

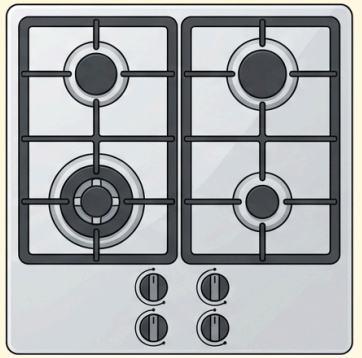
- **#1 Visibility of System Status:**
  - Keep users informed of current state within reasonable latency (`100ms` button press, `1s` transition, `10s` `%` progress bar + live order steps).
- **#2 Match Between System & Real World:**
  - Speak the **user's language** and follow real-world spatial mappings (see the classic **Stove Burner Mapping** on the right!)—never leak SQL/RPC codes.
- **#3 User Control & Freedom:**
  - Always provide an **"Emergency Exit"** (`Undo`, `Redo`, `Cancel Order within 60s`, clear modal `×`) so users never feel trapped by an accidental click.

Unnatural Mapping (Arbitrary)



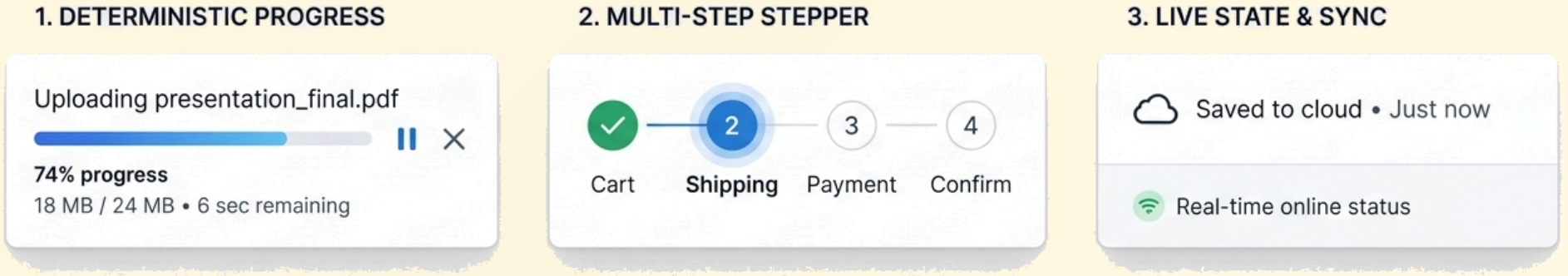
High Cognitive Friction — Which knob controls which burner? Requires labels and mental translation.

Natural Mapping (Spatial Correspondence)



Intuitive Operation — Subconscious, error-free physical mapping.

Visual Examples: Heuristic #1 Visibility of System Status



Nielsen's Heuristics #4 – #7: Consistency, Errors, Memory & Speed

> ***Cluster 2 (Cognitive Efficiency & Error Prevention):*** *Leverage conventions, prevent* ***Slips & Mistakes*** *before they happen, minimize memory load, and provide* ***Accelerators*** *for experts.*

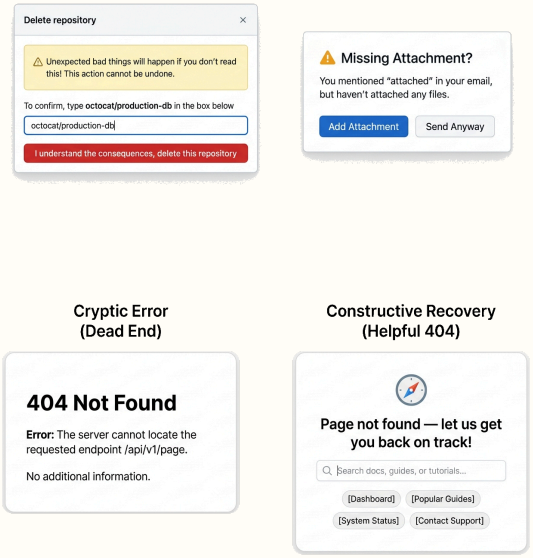
Heuristic (`#4 – #7`)	Core Principle & Cognitive Rationale	Software UI Engineering Examples (`[Good]` vs. `[Violation]`)
#4 Consistency & Standards	Users should not have to wonder whether different words, situations, or actions mean the same thing (**Internal & External Consistency**).	**[Good]**: Using standard platform navigation and consistent button terminology (`"Place Order"` everywhere). **[Violation]**: Calling the cart `"Basket"` on screen 1, `"Tray"` on screen 2, and `"Bag"` on screen 3.
#5 Error Prevention	Better than a good error message is a **careful design that prevents a problem from occurring** in the first place (eliminate **Slips** & **Mistakes**).	**[Good]**: Graying out past calendar dates; Gmail warning "*You wrote 'attached' but attached no file.*" **[Violation]**: Letting users type `"Feb 31"` in a plain text box and failing only after form submission.
#6 Recognition Rather Than Recall	Minimize user memory load by making **objects, actions, and options visible**; users should not have to remember info from one screen to another.	**[Good]**: Showing recent delivery addresses, dish thumbnails, and inline promo-code selector. **[Violation]**: Forcing users to memorize a 10-digit order ID from Page A to type into Support Page B.

Nielsen's Heuristics #8 – #10 & Visual Error Handling

1. Heuristics #8, #9 & #10 (Signal & Recovery)

- **#8 Aesthetic & Minimalist Design (Signal-to-Noise)**:
  - Remove irrelevant marketing clutter that competes with core task controls (Edward Tufte's *Data-Ink Ratio*).
- **#9 Recognize, Diagnose & Recover from Errors**:
  - **3 Rules of Error Messages**: *(a)* Plain human language (no SQL/hex codes), *(b)* Precise diagnosis, *(c)* **Constructive recovery link/button** without wiping user input!
- **#10 Help & Documentation**:
  - Provide concise, contextual inline tooltips (`[?]`) and interactive onboarding spotlights focused on user tasks.

2. Visual Comparison: #5 Prevention vs. #9 Recovery



Cognitive Walkthrough: Task-Focused Analytical Inspection

> ***Complementing Heuristic Evaluation (Wharton, Rieman, Lewis & Polson, 1994):*** *While Heuristic Evaluation audits a UI broadly against 10 principles, a* ***Cognitive Walkthrough*** *steps through a specific* ***task action-by-action*** *to evaluate* ***first-time Learnability****.*

1. The 4 Standard Questions Asked at Every Action Step

- For each micro-step `k` in a novice user's task sequence, evaluators ask **4 cognitive questions** (mapped to Norman's Action Cycle!):
  1. **Goal Formation**: Will the user realistically **try to achieve the right effect** at this step?
  2. **Visibility (Affordance)**: Will the user **notice that the correct UI control is available** on the screen?
  3. **Label Association (Signifier)**: Will the user **associate the control's label/icon with their intended goal**?
  4. **Feedback Evaluation**: After performing the action, will the user **see clear feedback confirming progress** toward their goal?

2. Heuristic Evaluation vs. Cognitive Walkthrough

Comparison Dimension	Heuristic Evaluation	Cognitive Walkthrough
Inspection Scope	Holistic screen-by-screen audit against **10 Heuristics**.	Step-by-step trace of **specific task flows**.
Primary Focus	Overall usability (`#1-#10`), consistency & error handling.	**First-time Learnability** & exploratory discovery.
Theoretical Root	Empirical usability principles (Nielsen, 1994).	**CE+ Theory of Exploratory Learning** & Norman's Action

Concept Check Question 5 (CCQ 5)

When a user attempts to register an account on a web portal, typing an already-registered email and clicking **"Submit"** clears the entire 12-field form and displays a banner reading: `"SQLSTATE[23505]: Unique violation on key idx_users_email"`. Which pair of **Jakob Nielsen's 10 Usability Heuristics** is most directly violated by this behavior?

- **A. #7 Flexibility and Efficiency of Use** and **#8 Aesthetic and Minimalist Design**.
- **B. #2 Match Between System and Real World** and **#9 Help Users Recognize, Diagnose, and Recover from Errors**.
- **C. #1 Visibility of System Status** and **#10 Help and Documentation**.
- **D. #4 Consistency and Standards** and **#6 Recognition Rather Than Recall**.

6.7 Empirical Usability Testing, Metrics & Ethical UX

> *"Testing one user early in the project is better than testing 50 near the end."*
> *— Steve Krug (Rocket Surgery Made Easy)*

Pioneers of Cognitive Science & Think-Aloud: Herbert A. Simon

- **Role & Distinctions:**
  - Professor of Computer Science & Psychology, **Carnegie Mellon University (CMU)**; recipient of both the **1975 ACM Turing Award** and the **1978 Nobel Prize in Economics**.
- **Foundational Contributions to HCI & UX Engineering:**
  - **Bounded Rationality & "*Satisficing*":** Proved that humans do not exhaustively scan every menu option for the mathematical optimum; they pick the **first reasonable option that seems "good enough"** (*explaining why users scan and click impulsively!*).
  - **The Attention Economy:** Formulated the law that information consumes human attention.
  - **The Think-Aloud Protocol (*Protocol Analysis*, with K. Anders Ericsson, 1984):** Established the rigorous cognitive methodology of having subjects verbalize Working Memory thoughts during problem-solving—now the **#1 usability testing method worldwide**.



Herbert A. Simon (1916–2001)
CMU / Nobel & ACM Turing Laureate

Formative vs. Summative Testing, Think-Aloud & Quantitative Metrics

> ***Iterative Discovery (`N = 5`) + Psychometric Benchmarking (`SUS & HEART`):*** *Pair qualitative* ***Think-Aloud*** *observation during design with quantitative metrics at release.*

1. Formative Think-Aloud (`N = 5` Poisson Model)

- **Formative ("*Find & Fix*" During Wireframing)**:
  - **Poisson Law (Nielsen & Landauer, 1993)**: `P = 1 - (1 - L)^n` (with `L ≈ 0.31`, **`n = 5` users uncovers `~85%`** of usability flaws!).
  - Run **3 iterative rounds × 5 users** (`v1 → fix → v2 → fix → v3`) rather than 1 giant round of 15 users!
- **3 Moderator Rules for Concurrent Think-Aloud**:
  1. Reassure: "*We are testing the software, never you.*"
  2. Use **non-leading task scenarios** (never repeat the button's exact text label in your prompt!).
  3. Watch **actions** over polite verbal compliments.

2. Quantitative Metrics, SUS (`0–100`) & HEART

- **Behavioral Task Metrics (Summative `N = 20–30+`)**:
  - **Task Completion Rate (`%`)**, **Time-on-Task (`s`)**, **First-Click Accuracy (`%`)**, and **Lostness (`L > 0.5` = lost)**.
- **Standardized Psychometric Scales**:
  - **SEQ (Single Ease Question)**: `1–7` rating right after each task (benchmark `>= 5.5`).
  - **SUS (System Usability Scale — John Brooke, 1986)**: 10 items scored `0–100` (percentile rank: `68` **is 50th percentile average**; `>= 80.3` **is Grade A** / Top 10%).
- **Production Scale**: **A/B Testing** (with **Guardrail Metrics**) + **Google HEART** (`Happiness, Engagement, Adoption, Retention, Task Success`).

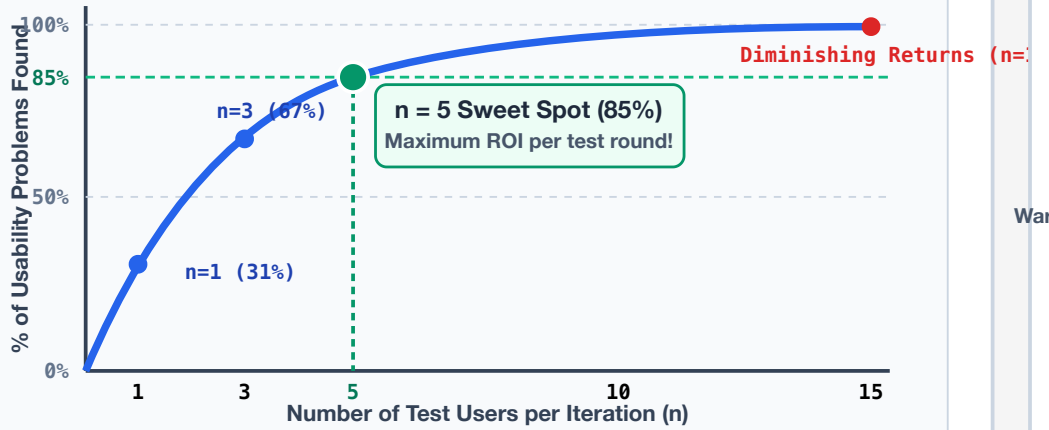
Conceptual Diagram: Nielsen's N=5 Curve, SUS Scale & HEART

Empirical Usability Engineering: Nielsen's N = 5 Curve & The SUS Benchmark Scale

Why 5 Formative Users Uncover ~85% of Usability Flaws (Left) & Interpreting System Usability Scale Scores 0–100 (Right)

1. NIELSEN & LANDAUER (1993) POISSON DISCOVERY MODEL

`P(Problems Found) = 1 − (1 − L)^n (where avg L ≈ 0.31)`



How to Budget 15 Test Participants Across a Project:

- [Anti-Pattern] 1 Big Round × 15 Users on Prototype v1:
  Users #6–#15 repeatedly hit the exact same obvious bugs; 0 users test your fixes!
- [Best Practice] 3 Iterative Rounds × 5 Users (v1 → Fix → v2 → Fix → v3):
  Uncovers 85% of flaws in v1, verifies the fixes in v2, and polishes edge cases in v3!

`* Exception: Summative benchmarking requires n = 20–30+ for tight statistical CIs.`

2. SYSTEM USABILITY SCALE (SUS) & GOOGLE HEART METRICS

John Brooke's (1986) 10-Item Scale: Scored 0–100 (Percentile Benchmark, NOT a Percentage!)

Interpreting Your Product's SUS Score (Sauro & Lewis Curved Grading Scale)

`★ SUS = 68 (50th Percentile Industry Avg)`

– 51 : Grade F (Unacceptable	51–68 (D/C)	68–80 (B)	80.3+ : Grade A

`High Churn / Detractors` — `Top 10% (NPS Promoters!)`

Warning: Because SUS is a curved percentile scale, a score of 60 is NOT "passing"—it is below average (Grade D)!

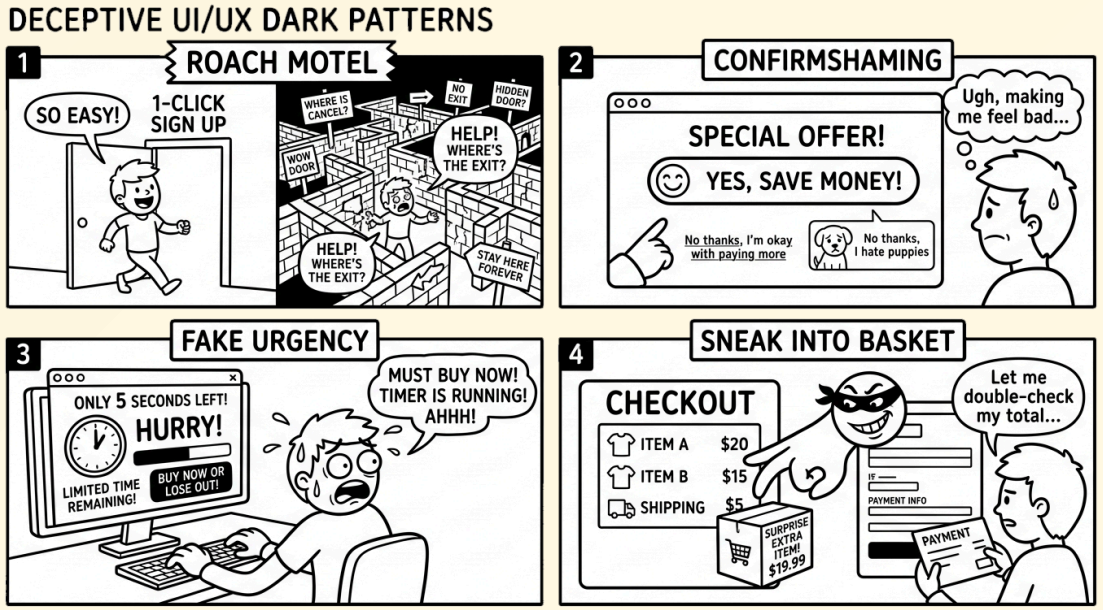
GOOGLE'S HEART FRAMEWORK (Goals → Signals → Metrics at Scale)

`H – Happiness:`	Subjective attitude (SUS score, post-order CSAT, App Store rating)
`E – Engagement:`	Depth & frequency of voluntary use (Orders/week, DAU/MAU ratio)
`A – Adoption:`	New users adopting a feature (% trying Group Order within 14 days)
`R – Retention:`	Cohort survival over time (D7 / D30 / D90 repeat buyers; inverse of Churn)
`T – Task Success:`	Behavioral Effectiveness & Efficiency (Completion %, Time-on-Task, Lostness L)

A/B Testing Rule: Always pair Primary Conversion Metrics with Guardrail Metrics
(e.g., monitor Cancel/Refund Rate & Support Tickets to prevent accidental-click Dark Patterns!)

Ethical UX Engineering: Dark Patterns vs. Transparent Design

- **What Are Dark Patterns? (Harry Brignull, 2010):**
  - Deceptive UI designs that exploit cognitive biases (**Default Bias, Sunk Cost, Visual Misdirection**) to trick users into unintended actions.
- **4 Classic Dark Patterns to Ban in Code Reviews:**
  - **1. Roach Motel**: 1-click sign-up, but canceling requires a phone call and 8 retention screens.
  - **2. Confirmshaming**: Guilt-tripping decline copy ("*No, I prefer wasting money*").
  - **3. Sneak into Basket / Drip Pricing**: Pre-checked add-ons or hiding mandatory fees until the final step.
  - **4. Visual Misdirection**: Inverting button colors so users accidentally click `"Keep Subscription"`.
- **Legal Mandate**: Banned by **EU Digital Services Act (DSA Art. 25)**, **GDPR**, and **FTC Click-to-Cancel** rules!



Peer Discussion 6.7: Designing a Usability Test Plan & Ethical Audit

Empirical UX Lab: Testing & Auditing "FoodPass" Subscription (4 Mins)

- **Scenario:** The business team wants to launch **FoodPass** (a $9.99/month free-delivery membership) inside our Food Delivery App.
- **Part 1 — Formative & Summative Usability Test Plan:**
  - You have **2 weeks** before launch. How many participants (`N`) will you recruit for your **Formative Think-Aloud** wireframe rounds vs. your final **Summative Benchmark**?
  - Write **1 concrete, non-leading Task Scenario** (without using the exact button label `"Manage FoodPass"`) to test whether users can pause or cancel their membership, and list **2 Quantitative Metrics** you will record.
- **Part 2 — Dark Pattern Ethical Audit:**
  - A PM proposes: *(a)* Pre-checking `"Add FoodPass ($9.99/mo)"` at checkout by default, and *(b)* Requiring users to go through 5 screens where `"Keep FoodPass"` is bright green and `"Cancel"` is a tiny gray link.
  - **Task:** Name the **2 Dark Patterns** present and rewrite the flow to achieve high ethical conversion without deception!



6.8 AI-Augmented UX Engineering, Synthesis & References

> *"In the era of Generative AI, the paradigm shifts twice: we use AI to accelerate UX workflows (AI for UX), and we invent new interaction paradigms for probabilistic AI systems (UX for AI)."*

The Dual Paradigm: "AI for UX" vs. "UX for AI" (Human-AI Interaction)

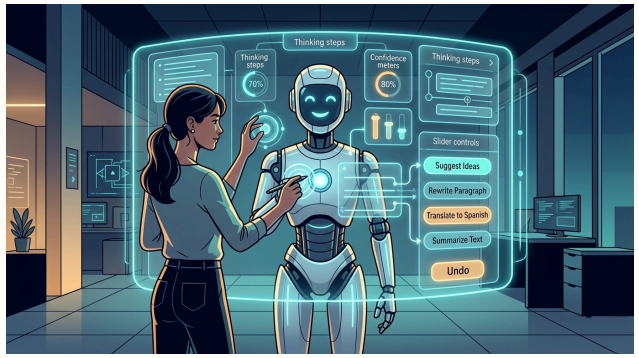
> ***Deterministic GUI vs. Probabilistic AI:*** *Traditional GUIs are* ***100% deterministic*** *(`Click Button A → State B`); Generative AI interfaces are* ***probabilistic and fallible****, requiring* ***Human-Centered AI (HCAI)*** *patterns!*

1. AI for UX (Copilot Across the HCD Lifecycle)



- **Research & Synthesis**: Clustering interview transcripts into **Affinity Diagrams** with verbatim quotes.
- **Prototyping & Pre-Audit**: Generating wireframe variants, domain microcopy, and running automated **WCAG / Heuristic pre-checks** (*never replacing real human testing!*).

2. UX for AI (Microsoft 18 HAI Guidelines, CHI '19)



- **1. Hybrid Intent UI (Defeat Blank-Box Paralysis)**: Pair chat prompts with **clickable starter chips & GUI sliders** (Heuristic #6 Recognition over Recall).
- **2. Source Grounding & Streaming (`< 400ms`)**: Show provenance citations and confidence indicators.
- **3. Human-in-the-Loop Agency**: Require explicit **Diff Preview + One-Click Undo** before AI executes actions!

Conceptual Recap: Fill-in-the-Blank Quiz (Part 1)

Word Bank: Gulf of Execution | Gulf of Evaluation | Fitts's Law | Extraneous Cognitive Load

1. In Don Norman's 7 Stages of Action, the gap between a user's intention and the actions allowed by the interface ("*How do I work this?*") is the [_____].
2. When a user clicks a button and cannot tell whether the system registered the action due to missing status feedback, they face a wide [_____].
3. In John Sweller's Cognitive Load Theory, unnecessary mental effort caused by cluttered layouts, inconsistent navigation, or cryptic jargon is called [_____].
4. The logarithmic predictive model $MT = a + b \cdot \log_2(D / W + 1)$ governing pointing time as a function of target distance and size is [_____].

Interactive Mode: Click any blank slot to reveal or hide the answer!

Conceptual Recap: Fill-in-the-Blank Quiz (Part 2)

Word Bank: Tree Testing | Recognition Rather Than Recall | Think-Aloud Protocol | Roach Motel

5. The empirical IA evaluation method ("reverse card sorting") that tests findability and first-click accuracy using a plain-text menu hierarchy is [_____].
6. Showing recent delivery addresses and visual autocomplete suggestions instead of forcing users to memorize codes applies Nielsen's Heuristic #6: [_____].
7. The formative usability testing method (Ericsson & Simon, 1984) in which participants continuously verbalize their thoughts while performing tasks is the [_____].
8. A deceptive Dark Pattern that makes subscribing take 1 click while burying cancellation behind phone calls and 8 retention screens is a [_____].

Interactive Mode: Click any blank slot to reveal or hide the answer!

Core Curriculum Synthesis & Classic Academic Literature

Section	Pioneer & Visual Concept Diagram	Golden UX Engineering Takeaway
6.1 Foundations	**Douglas Engelbart**; Garrett's 5 Planes & Morville Honeycomb SVG	UX is built bottom-up (`Strategy → Surface`); quantify Learnability, Efficiency, Memorability, Errors & Satisfaction.
6.2 Cognition	**Don Norman**; 7 Stages, 2 Gulfs, MHP & Cognitive Load SVGs	Narrow the **Gulfs of Execution & Evaluation**; align the **System Image** with the **User's Mental Model**; cut extraneous load.
6.3 Laws & a11y	**Ben Shneiderman**; Fitts's Law, Thumb Zones, Hick & Gestalt SVG	Maximize target `W`, minimize `D` & choice entropy `n`; absorb complexity in code; enforce `4.5:1` contrast & non-color cues.
6.4 HCD Research	Double Diamond, 2×2 Research Matrix & Journey Map SVGs	"*You are not the user!*" Triangulate behavioral & attitudinal research; trace Personas/JTBD/HTA directly to the **SRS**.
6.5 IA & Prototyping	Card Sorting, Tree Testing & Brad Frost Atomic Design SVG	Validate taxonomy findability first; sketch Lo-Fi to get the *right design*; scale UI via **Atoms** → **Organisms** & Tokens.
6.6 Heuristics	**Jakob Nielsen**; 10 Usability Heuristics (`0–4` Severity) & Stove Map	Run 3–5 independent inspections (`~80%` bug discovery); prevent Slips & Mistakes; audit novice steps via `Q1–Q4`.
6.7 Testing & Ethics	**Herbert A. Simon**; `N = 5` Poisson Curve, SUS `0–100` & HEART SVG	Test 5 users per round (**Think-Aloud**); benchmark via **SUS (`68` avg)**; reject **Roach Motel / Confirmshaming** Dark Patterns.
6.8 AI & UX	**AI for UX** vs. **UX for AI (Microsoft 18 HAI Guidelines)**	Pair natural language with GUI controls, ground sources, and require **Human-in-the-Loop Preview + Undo**.

- **Essential References:** Don Norman (*The Design of Everyday Things*); Jakob Nielsen (*Usability Engineering*); Ben Shneiderman (*Designing the User Interface*); Herbert A. Simon & K. Anders Ericsson (*Protocol Analysis*); Steve Krug (*Don't Make Me Think*); ISO 9241-210 & W3C WCAG 2.2

Appendix A: Suggested Answers for CCQ 1 – CCQ 5

Question	Ans	Core UX / HCI Principle & Concise Engineering Rationale
CCQ 1 (Sec 6.1) *Usability Dimensions*	**B**	**Efficiency vs. Learnability (Nielsen's 5 Components)**: Novice interns learn in 10 mins (high *Learnability*), but requiring 14 clicks across 5 modals for experienced daily nurses bottlenecks expert task speed (**Efficiency**).
CCQ 2 (Sec 6.2) *Norman's Two Gulfs*	**C**	**Wide Gulf of Evaluation**: The engineer executed the click easily (narrow *Gulf of Execution*), but the lack of immediate visual feedback (`< 100ms` button state / `< 1s` spinner) prevented perceiving the new system state, causing duplicate clicks.
CCQ 3 (Sec 6.3) *Fitts's Law (`D` & `W`)*	**B**	**Infinite Target Width (`W = ∞`) at Screen Edges**: Under `MT = a + b · log2(D/W + 1)`, pinning global menus flush against screen edges/corners uses the display boundary to stop cursor overshoot, dramatically lowering the Index of Difficulty.
CCQ 4 (Sec 6.5) *IA Empirical Testing*	**B**	**Tree Testing (Reverse Card Sorting)**: Evaluates an existing text hierarchy stripped of visual layout to measure **Findability Success Rate (%)** and **First-Click Directness (%)** prior to wireframing.
CCQ 5 (Sec 6.6) *Nielsen's 10 Heuristics*	**B**	**#2 Match Real World & #9 Error Recovery**: Exposing raw SQL error codes (`SQLSTATE[23505]`) violates **#2** (system-oriented jargon); wiping out 12 user-typed fields with no constructive fix violates **#9** (and **#5 Error Prevention**).

Appendix B: Suggested Solutions for Peer Discussions 6.3, 6.5 & 6.7

1. Solution — Peer Discussion 6.3 (Checkout Audit) & 6.5 (Group IA)

- **Peer Discussion 6.3 (4 Checkout Flaws & Fixes)**:
  1. **Fitts's Law + Heuristic #5**: Enlarge `"Place Order"` to full-width `52px` bottom bar (`↑ W`, `↓ D` in thumb zone); move `"Empty Cart"` into a top-right overflow menu with confirmation.
  2. **Hick-Hyman Law**: Replace 28 flat radio buttons with the user's **last-used payment method** as default + `3` top categories (`Card | Apple/Google Pay | Bank`).
  3. **Postel's & Tesler's Laws + Heuristic #9**: Auto-strip `-`/`()` spaces in JS; preserve all 8 address fields on validation error!
  4. **WCAG 2.2 Level AA (`1.4.1` & `1.4.3`)**: Increase contrast to `>= 4.5:1` (`#b91c1c` on `#fef2f2`) and add `[!] Required` icon + text below each missing field.
- **Peer Discussion 6.5 (Group Order IA & 4 States)**:
  - **IA Placement**: Place `"[+ Group Order]"` button directly in the **Restaurant Menu Header** (matches user mental model when browsing a restaurant).
  - **Edge State Override**: Show per-member status (`Adding...` vs. `Ready ✓`), auto-cap items at `$25` with inline split-pay option, and give Host a **"Lock Cart & Checkout Now"** override so 1 away coworker never blocks the order!

2. Solution — Peer Discussion 6.7 (Usability Test & Ethics)

- **Part 1 — Test Plan & Non-Leading Scenario**:
  - **Sample Plan**: Run **2 formative rounds × 5 users (`N = 10`)** using **Think-Aloud** on Figma prototypes to fix flows, then **1 summative round (`N = 25`)** to benchmark.
  - **Non-Leading Prompt**: "*Imagine you subscribed to FoodPass last month, but you are traveling abroad for 3 weeks and want to stop being billed next month. Show me what you would do.*" (Never say the word `"Manage"` or `"Cancel"`!).
  - **Metrics**: **Task Completion Rate (%)**, **Time-on-Task (s)**, **First-Click Accuracy (%)**, and post-task **SEQ / SUS**.
- **Part 2 — Dark Pattern Audit & Ethical Redesign**:
  - **Violations**: `(a)` **Sneak into Basket** (pre-checked opt-out box); `(b)` **Roach Motel + Visual Misdirection** (5 retention screens with inverted button colors).
  - **Ethical Fix**: Show transparent math ("*Save $4.50 on this order with FoodPass — [Add Free Trial]*" explicit Opt-In) and provide a symmetric **1-Click Pause / Cancel** button in Account Settings!